

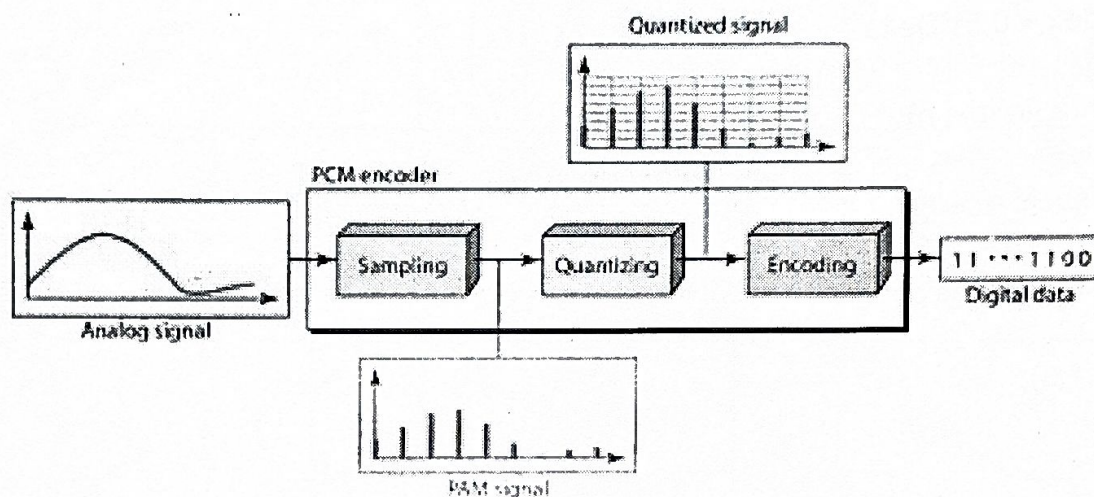
Title: Analog to Digital Conversion

Objectives:

1. To understand the steps in analog to digital conversion with given example 2. To simulate the analog to digital conversion steps in Simulink and analyze the results

OUTCOME: apply concept of sampling for reliable communication

Theory:



Problem Statement:

- Assuming that a 3-bit unipolar ADC channel accepts analog input (sine wave) ranging from 1 to -1 volts with frequency of 2 Hz. Write a program to plot sampled signal, quantized signal, and encoded signal. Show all background mathematical calculations.
- Design a model in Simulink for the above statement.

Expected Output:

- Various calculations to design Simulink model
- Plot of sampled, quantized and encoded signal

Matlab Code:

```
clc;
clear;
close all;
% Given parameters
fm = 2;           % Frequency of sine wave in Hz
A = 1;           % Amplitude in volts
fs = 20;         % Sampling frequency in Hz
n = 3;           % Number of ADC bits
L = 2^n;         % Number of quantization levels
% Time vectors
t = 0:0.001:1;
```

calculations?


```

ts = 0:1/fs:1;
% Analog signal
x = A*sin(2*pi*fm*t);
% Sampled signal
xs = A*sin(2*pi*fm*ts);
% ADC voltage range
Vmin = -1;
Vmax = 1;
% Quantization step
Delta = (Vmax - Vmin)/L;
% Quantization
q_index = floor((xs - Vmin)/Delta);
% Limit the index
q_index(q_index < 0) = 0;
q_index(q_index > L-1) = L-1;
% Quantized voltage
xq = Vmin + (q_index + 0.5)*Delta;
% Binary encoding
encoded = dec2bin(q_index, n);
% Display calculations
fprintf('Number of levels = %d\n', L);
fprintf('Quantization step = %.2f V\n', Delta);
% Plot analog and sampled signal
figure;
plot(t,x,'LineWidth',1.5);
hold on;
stem(ts,xs,'filled');
grid on;
xlabel('Time (s)');
ylabel('Amplitude (V)');
title('Sampled Signal');
% Quantized signal
figure;
stairs(ts,xq,'LineWidth',1.5);
hold on;
stem(ts,xs,'filled');
grid on;
xlabel('Time (s)');
ylabel('Amplitude (V)');
title('Quantized Signal');
% Quantization index
figure;
stem(ts,q_index,'filled');
grid on;
xlabel('Time (s)');
ylabel('Quantization Level');
title('Encoded/Quantization Levels');
% Display encoded data
disp('Encoded 3-bit data:');
disp(encoded);

```


Matlab Output:

Figure 1 × Figure 2 × Figure 3 ×

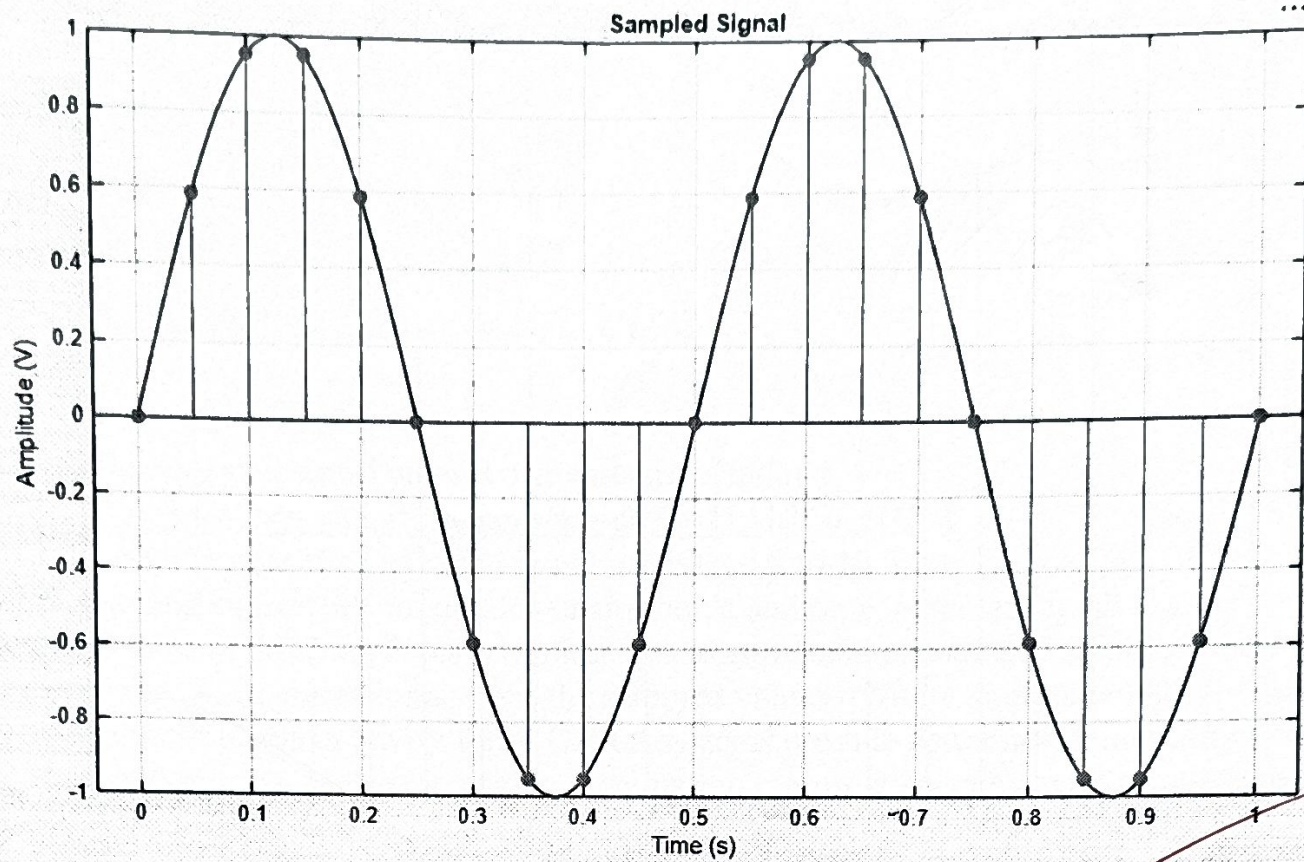
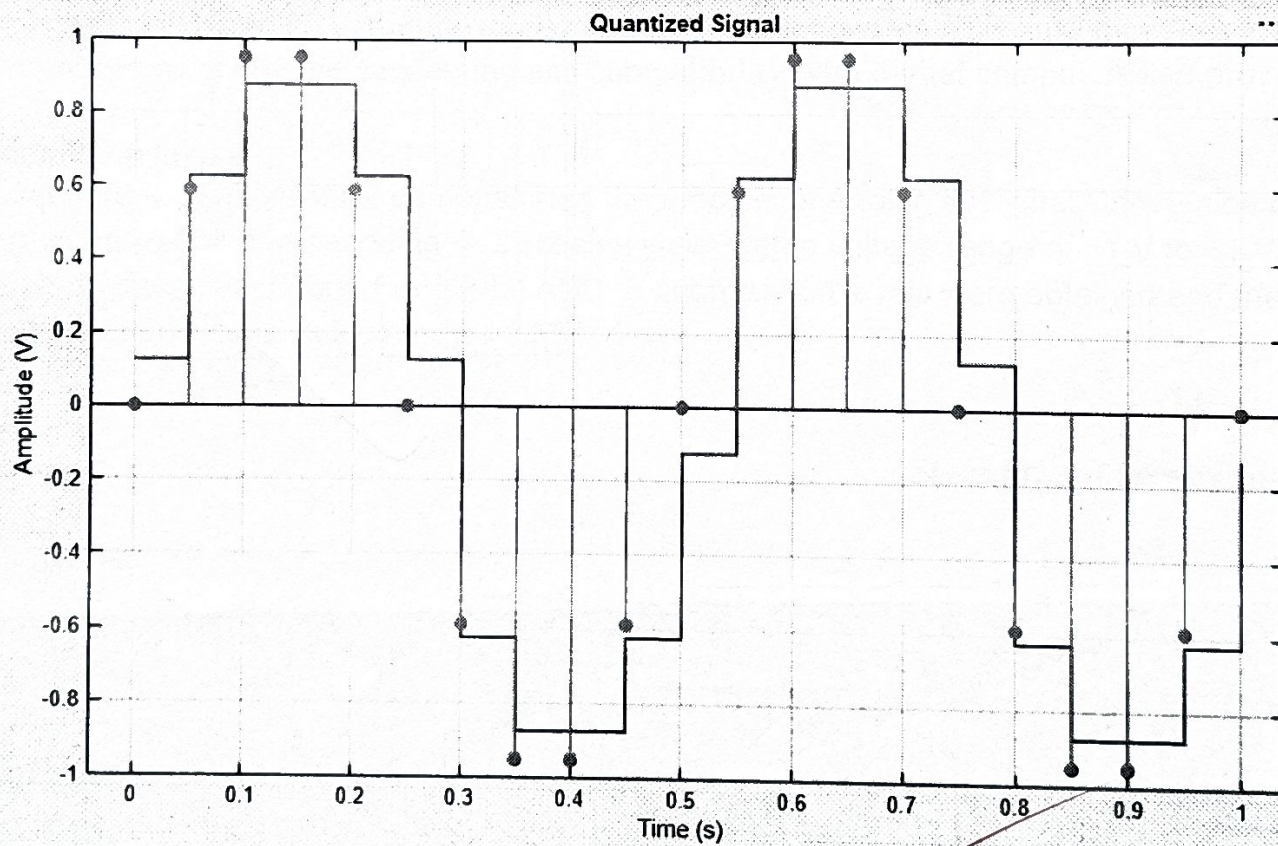
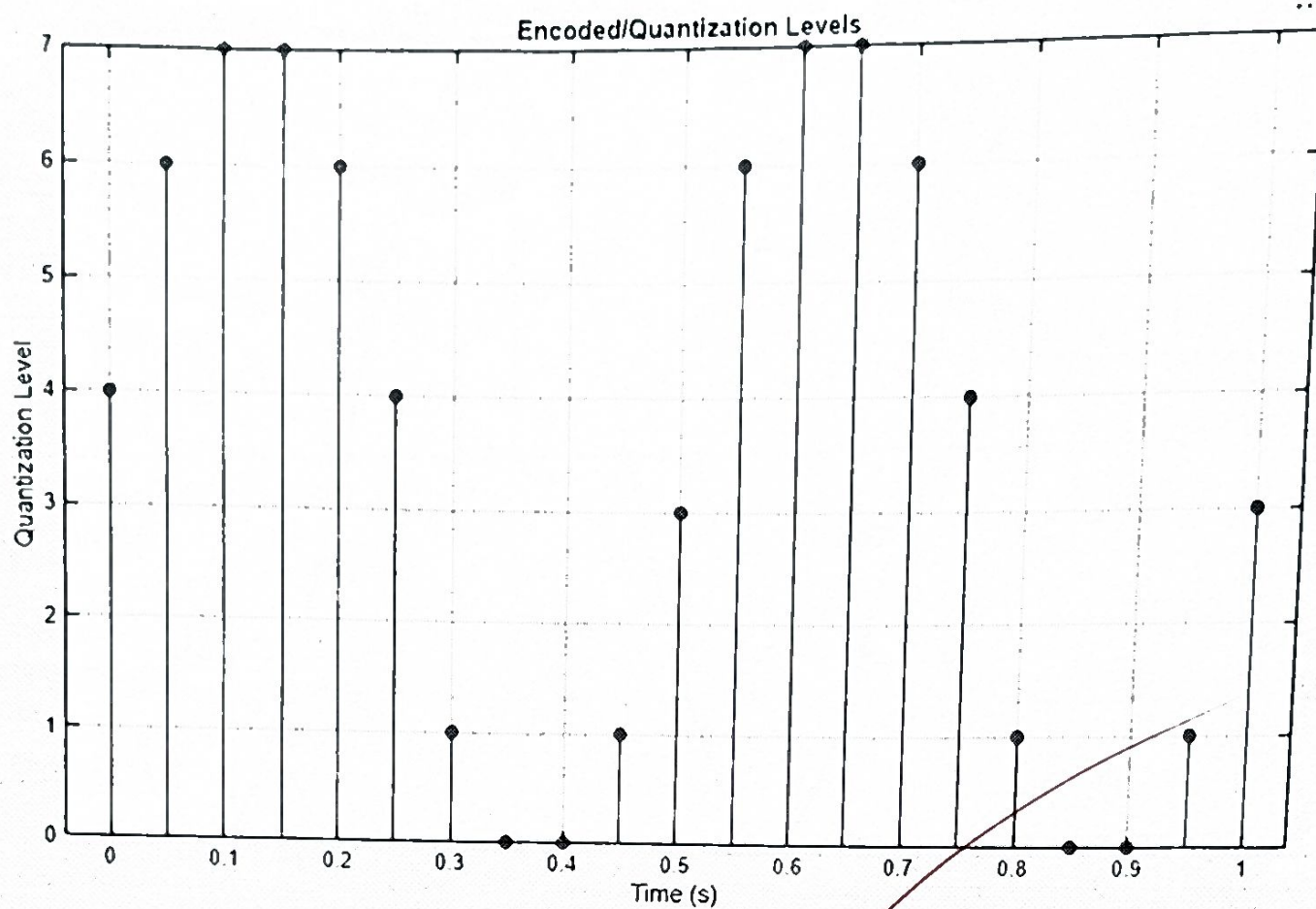


Figure 1 × Figure 2 × Figure 3 ×





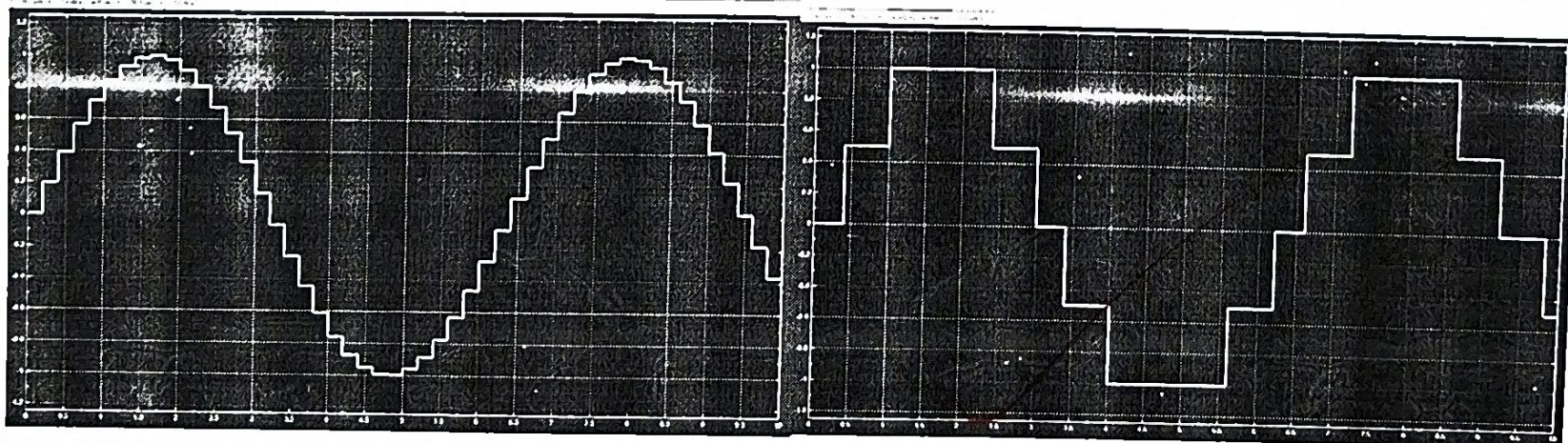
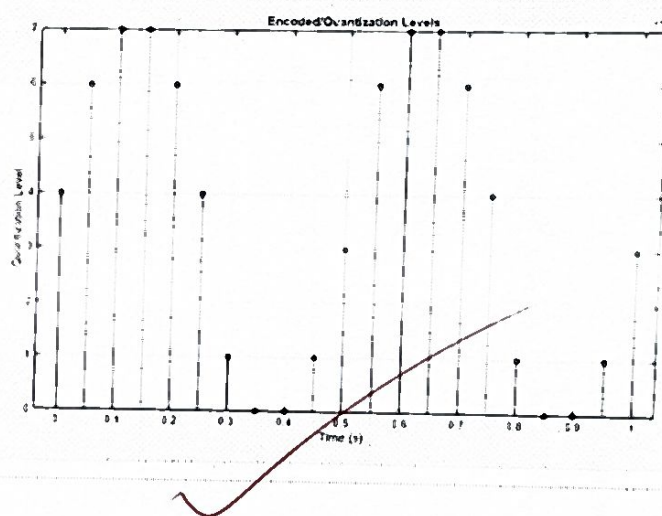
enc_quant_1.m, 1671612, 04/04/04
 Q12, 42, 0.000000, 0.000000, 0.000000, 0.000000

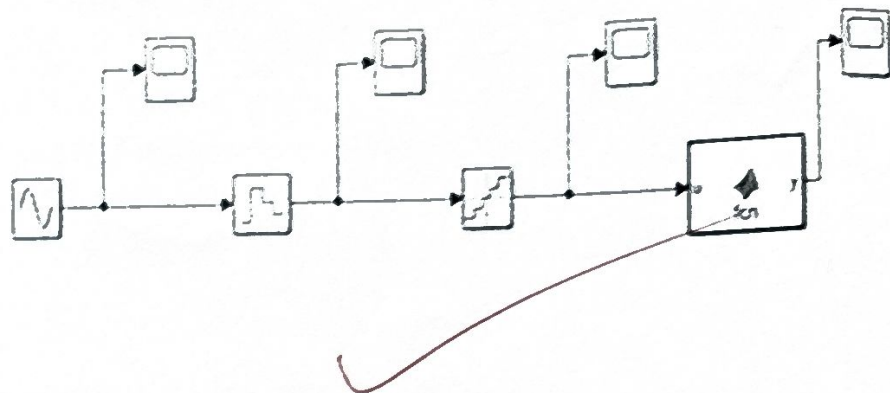
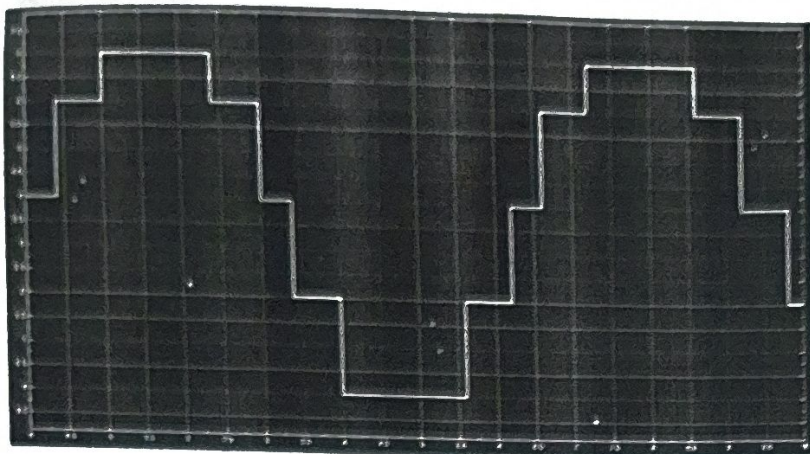
```

1 clear;
2 close;
3 close all;
4
5 % Given parameters
6 fs = 1; % Sampling frequency (Hz)
7 N = 1000; % Number of samples
8 Ts = 1/fs; % Sampling period (s)
9 T = Ts*N; % Total time (s)
10
11 % Time vectors
12 t = 0:Ts:(T-Ts); % Time vector (s)
13 ts = 0:Ts:(T-Ts); % Time vector (s)
14
15 % Input signal
16 x = 2*cos(2*pi*10*t); % Input signal (V)
17
18 % Sampled signal
19 xs = 2*cos(2*pi*10*ts); % Sampled signal (V)
20
21 % ADC voltage range
22 Vmin = -1; % Minimum voltage (V)
23 Vmax = 1; % Maximum voltage (V)
24
25 % Quantization step
26 Delta = (Vmax - Vmin)/8; % Quantization step (V)
27
28 % Quantization
29
  
```

Command Window
 Quantization step = 0.25 V
 Encoded 9-bit data:
 100
 110
 100
 111
 110

Figure 1.8, Figure 2, Figure 3





Activity

Watch the following video and answer the questions below:

o https://www.youtube.com/watch?v=I1AhW6_HQpI

o Write reflection of video in not more than 8 to 10 lines.

I learned that analog signals are continuous in amplitude and time, whereas digital signals are discrete in both. The process of ADC involves sampling, quantization and encoding. Sampling determines the discrete time instants. Quantization assigns the sampled values to finite amplitude levels. Encoding then represents these levels in binary form. Digital systems provide better noise immunity and are easier to store and process. However, analog systems can represent signals continuously without quantization. Overall, digital systems are more suitable for modern communication and signal-processing applications.

o According to you, which system is better? Analog or Digital?

Digital systems are generally better for modern communication systems because they provide better noise immunity, easier storage, processing and compatibility with digital computers and processors.

Discussion/Conclusion

In this experiment, I successfully demonstrated the process of Analog to Digital Conversion using sampling, quantization and encoding. A 2 Hz sine wave with a voltage range of -1 V to $+1\text{ V}$ was considered as the analog input. For a 3-bit ADC, 8 quantization levels were obtained and the quantization step size was calculated as 0.25 V .

Sh 11/9/25
Signature of faculty in-charge